

A Quick Proof of Global Regularity for Three-Dimensional Navier–Stokes

The datum-generated compatible intersection as an estimate generator

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Abstract

Fix one smooth divergence-free datum and one positive parameter ν . They generate one maximal velocity–pressure history. The equation generates the complete mixed derivative jet of that history, and the datum-generated whole-terminal theorem places every finite block of the jet in a compatible Sobolev rectangle. The projective intersection of those rectangles is one all-orders object, not a collection of unrelated estimates. Its coordinate projections give every finite-order continuation bound, a common H^∞ endpoint, and the complete pressure-compatible endpoint jet. Smooth local existence and uniqueness then restart the same history across any proposed finite terminal time. The maximal time is infinite.

Theorem 1 (Global regularity). *Let $\nu > 0$ and let*

$$u_0 \in H^\infty(\mathbb{R}^3; \mathbb{R}^3), \quad \nabla \cdot u_0 = 0.$$

The three-dimensional incompressible system on $\mathbb{R}^3$ has one global smooth velocity–pressure history generated by u_0 , with pressure fixed by the Riesz-transform normalization. In particular,

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

The proof carries one object through the following chain:

$$\boxed{\begin{aligned} u_0 &\longrightarrow (u, p) \longrightarrow \{J_{n,\alpha}, \Pi_{n,\alpha}\}_{n,\alpha} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M} \\ &\longrightarrow \mathcal{I}[u_0] \longrightarrow (u_*, \{V_n^*, Q_n^*\}_{n \geq 0}) \longrightarrow T_* = \infty. \end{aligned}} \quad (1)$$

The point of the notation is custody: every arrow still refers to the same history generated by the same datum.

1 One pressure-complete history

Let (u, p) be the maximal classical history generated by u_0 , defined on $I_* = [0, T_*)$. It satisfies

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0. \quad (2)$$

Incompressibility and the chosen normalization determine pressure from this same velocity field:

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (3)$$

Assume for contradiction that $T_* < \infty$. Nothing below changes the datum, parameter ν , pressure convention, or history.

Put

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

Differentiating (3) and (2) gives, for every $n \in \mathbb{N}_0$,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (4)$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \quad (5)$$

The equation does not generate a new velocity field at height n . It reads a higher response of the same field.

For a spatial multi-index α , define the pressure-complete mixed jet

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

Leibniz' rule turns (2) into

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}. \end{aligned} \quad (6)$$

The temporal tower, the spatial Sobolev tower, pressure, vorticity, and every later continuation readout are coordinates of this one mixed jet.

2 Datum-generated rectangles and their intersection

For $N \in \mathbb{N}_0$, $M \geq 1$, and $0 < T < T_*$, define the finite rectangle norm

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) &:= \sum_{n=0}^N \|\partial_t^n u\|_{L^2(0,T;H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|\partial_t^{n+1} u\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (7)$$

The rectangle $\mathcal{R}_{N,M}[u_0]$ is the corresponding finite restriction of the pressure-explicit jet (6).

Theorem 2 (Datum-generated whole-terminal compatible rectangles). *The datum u_0 , parameter ν , and their one maximal velocity–pressure history generate, for every finite (N, M) , a rectangle $\mathcal{R}_{N,M}[u_0]$ satisfying the whole-terminal estimate*

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) \leq C_{N,M}(u_0, \nu, T_*) < \infty. \quad (8)$$

The constant may depend on the finite coordinate (N, M) ; no uniformity in all orders is needed. If $N' \leq N$ and $M' \leq M$, restriction obeys

$$\rho_{N',M'}^{N,M}(\mathcal{R}_{N,M}[u_0]) = \mathcal{R}_{N',M'}[u_0]. \quad (9)$$

Every shared coordinate is the corresponding derivative of the same history.

Finite-coordinate realization. The pressure-explicit adjacent-row theorem supplies, for every finite (n, m) ,

$$\partial_t^n u \in L^2(0, T_*; H^{m+1}), \quad \partial_t^{n+1} u \in L^2(0, T_*; H^{m-1}),$$

with the datum-generated bound uniform over all truncations $T < T_*$. Fix (N, M) . Adding the finitely many squared row norms gives (8). Recurrences (4)–(6) identify every row with the actual derivatives of (u, p) . Forgetting higher time coordinates and applying the lower Sobolev embeddings cannot change a shared derivative, which gives (9). $\square$

Compatibility is what permits the finite pieces to become one object. Define

$$\mathcal{I}[u_0] := \varprojlim_{N, M} \mathcal{R}_{N, M}[u_0]. \quad (10)$$

Proposition 3 (Projective assembly). *The velocity realization of $\mathcal{I}[u_0]$ is the same u in every coordinate, and*

$$u \in \mathcal{X}(0, T_*) := \bigcap_{r, m \in \mathbb{N}_0} H^r(0, T_*; H^m(\mathbb{R}^3)). \quad (11)$$

Its zeroth temporal projection is

$$\pi_0 \mathcal{I}[u_0] = u \in \mathcal{S}_0(0, T_*) := \bigcap_{m \in \mathbb{N}_0} L^2(0, T_*; H^m(\mathbb{R}^3)). \quad (12)$$

Proof. Fix $r, m \in \mathbb{N}_0$. Choose $N \geq r$ and $M \geq \max\{1, m\}$. The rectangle estimate places $\partial_t^n u$ in $L^2(0, T_*; H^m)$ for $0 \leq n \leq r$, so $u \in H^r(0, T_*; H^m)$. Since (r, m) was arbitrary, (11) follows. Equation (9) says that all lower-coordinate representatives agree, giving one inverse-limit element and the projection (12). $\square$

The order is essential: the complete compatible intersection now exists. The familiar critical and continuation estimates are its projections.

3 The intersection reaches the terminal time

Proposition 4 (Master trace and embedding readout). *For every finite r, m , the intersection gives*

$$\partial_t^r u \in C([0, T_*]; H^m), \quad M_{r, m} := \sup_{0 \leq t \leq T_*} \|\partial_t^r u(t)\|_{H^m} < \infty. \quad (13)$$

If $k \in \mathbb{N}_0$, $2 \leq p \leq \infty$, and

$$m > k + \frac{3}{2} - \frac{3}{p},$$

then for every $1 \leq q \leq \infty$,

$$\left\| \nabla^k \partial_t^r u \right\|_{L^q(0, T_*; L^p)} \leq C_{k, m, p} T_*^{1/q} M_{r, m}, \quad (14)$$

where $T_^{1/\infty} = 1$.*

Proof. Choose the coordinate $(r+1, m)$ in (11). Then $\partial_t^r u \in H^1(0, T_*; H^m)$, and the Hilbert-valued one-dimensional trace theorem gives (13). The spatial Sobolev embedding gives

$$\left\| \nabla^k \partial_t^r u(t) \right\|_{L^p} \leq C_{k, m, p} \|\partial_t^r u(t)\|_{H^m}.$$

Taking the time norm proves (14). $\square$

There is also a direct endpoint payment contained in the zeroth row alone. Write the nonlinear term with the Leray projection:

$$\mathcal{B}(u, u) := (u \cdot \nabla)u + \nabla p = \mathbb{P}\nabla \cdot (u \otimes u).$$

For every finite m ,

$$\|\mathcal{B}(u, u)\|_{H^m} \leq C_m \|u\|_{H^{m+2}}^2. \quad (15)$$

By (12), the right side is integrable on $(0, T_*)$. Also $\Delta u \in L^1(0, T_*; H^m)$, since the interval is finite and $u \in L^2(0, T_*; H^{m+2})$. Equation (2) now gives

$$u_t \in L^1(0, T_*; H^m), \quad u \in W^{1,1}(0, T_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (16)$$

All endpoint values are traces of the same velocity field. They define one common endpoint

$$u_* := u(T_*) \in \bigcap_{m \in \mathbb{N}_0} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (17)$$

Pressure reaches the endpoint with the velocity. For every n , the trace from Proposition 4 gives

$$V_n^* := \lim_{t \uparrow T_*} \partial_t^n u(t) \in H^\infty.$$

Starting with $V_0^* = u_*$, set

$$Q_n^* := R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^* V_{n-a,j}^*, \quad (18)$$

$$V_{n+1}^* := \nu \Delta V_n^* - \sum_{a=0}^n \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^*. \quad (19)$$

Multiplication, differentiation, and the Riesz transforms are continuous on the required Sobolev levels. Passing $t \uparrow T_*$ through (4)–(5) identifies (18)–(19) with the actual left endpoint traces. Thus

$$V_n^*, Q_n^* \in H^\infty \quad \text{for every } n \in \mathbb{N}_0. \quad (20)$$

The terminal object is the complete pressure-compatible jet of the original history.

4 Same-history restart

Smooth local existence from $u_* \in H^\infty$ produces a solution on $[T_*, T_* + \varepsilon)$. Its right jet is generated by (18)–(19), so it agrees with the left jet in (20). Local uniqueness on the overlap glues the restart to the original history. The resulting smooth solution extends past T_* , contradicting finite maximality. Hence

$$T_* = \infty. \quad (21)$$

Pressure is recovered from (3) at every time, and the mixed recurrence propagates all space–time derivatives. This proves Theorem 1.

The proof has now completed the chain in its original order:

$$\begin{aligned} u_0 &\longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M} \longrightarrow \mathcal{I}[u_0] \longrightarrow u_* \in H^\infty \\ &\longrightarrow \text{pressure-complete endpoint jet} \longrightarrow \text{same-history restart} \longrightarrow T_* = \infty. \end{aligned} \quad (22)$$

5 Classical estimates as coordinate projections

The closure above does not require a separate continuation criterion. Once $\mathcal{I}[u_0]$ is available, each fixed-order criterion follows by choosing a sufficiently high coordinate in Proposition 4. Three familiar certificates are immediate:

$$u \in L^2(0, T_*; L^\infty), \quad u \in L^\infty(0, T_*; L^3), \quad \omega \in L^1(0, T_*; L^\infty). \quad (23)$$

Indeed, use respectively $u \in L^2 H^2$, $u \in C_t H_x^1$, and $u \in L^2 H^3$, followed by Sobolev embedding and Cauchy–Schwarz in time.

Corollary 5 (Finite-order continuation catalogue). *Every fixed finite-order classical readout in the left column is finite for the reason in the right column.*

Readout	Projection from $\mathcal{I}[u_0]$
Serrin class	$u \in L_t^2 H_x^2 \hookrightarrow L_t^2 L_x^\infty$, with $2/2 + 3/\infty = 1$.
Critical endpoint	$u \in C_t H_x^1 \hookrightarrow L_t^\infty L_x^3$.
Vorticity, strain, and Lipschitz control	$u \in L_t^2 H_x^3$ gives $\omega, S, \nabla u \in L_t^2 L_x^\infty \subset L_t^1 L_x^\infty$.
Fixed-order dyadic shells	For inhomogeneous blocks Δ_j , Bernstein and Cauchy–Schwarz give $\sum_j \int_0^{T_*} \ \Delta_j \omega\ _\infty dt \lesssim T_*^{1/2} \ u\ _{L_t^2 H_x^3}$.
Pressure and pressure gradient	The Riesz formula (3), Sobolev products, and higher coordinates give every finite mixed norm required for p and ∇p .
Type-I quantities	Uniform finite-order bounds imply $(T_* - t)^\gamma \ \nabla^k u(t)\ _\infty \rightarrow 0$ for each fixed k and $\gamma > 0$.
Critical-height readout	$H(t) = \frac{1}{2} \ \Lambda^{1/2} u(t)\ _2^2$ is continuous and bounded; its finite-time octave count is finite.
Ultraviolet tail	For $H_{>K}(t) = \frac{1}{2} \int_{ \xi >K} \xi \widehat{u}(t, \xi) ^2 d\xi$, higher uniform Sobolev control gives $\sup_{t \leq T_*} H_{>K}(t) \rightarrow 0$.
Partial-regularity readout	The smooth representative supplied above has an empty singular set.

These are fixed-order, finite-horizon consequences. A sum over every Sobolev order also needs growth control, and infinite-horizon shell integrability also needs decay. Small-data thresholds, symmetry classes, and vorticity-direction conditions keep their own assumptions. The list is not a second route: each entry is a finite projection of the one datum-generated object that already reaches the endpoint.

References

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